Unsupervised 3-D Tensor Subspace Decomposition Network for Spatial–Temporal–Spectral Fusion of Hyperspectral and Multispectral Images

Weiwei Sun, *Senior Member, IEEE*, Kai Ren, *Student Member, IEEE*, Xiangchao Meng, *Member, IEEE*, Gang Yang, Jiangtao Peng, *Senior Member, IEEE*, and Jiancheng Li

***Abstract*— Due to sensor design limitations and the influence of weather factors, it is currently challenging to obtain remote sensing images with high temporal, spatial, and spectral resolution. Spatial–temporal–spectral fusion aims to integrate the temporal, spatial, and spectral information from multiple sources of remote sensing images to reconstruct a remote sensing image with high temporal, spatial, and spectral resolution. Existing methods typically require at least three types of data to achieve spatial–temporal–spectral fusion. However, acquiring remote sensing data observed at the same time poses significant difficulties. The major challenge lies in effectively utilizing hyperspectral images (HSIs) with low spatial and temporal resolution and multispectral images with high temporal and spatial resolution to reconstruct remote sensing images with high temporal, spatial, and spectral resolution. To address the aforementioned issues, we propose a novel unsupervised 3-D tensor subspace decomposition network. Our method incorporates the theory of 3-D tensor subspace decomposition, utilizing a 3-D hyperspectral/multispectral tensor subspace extraction network to predict the hyperspectral tensor subspace features with low spatial resolution missing at other times (to better understand, the missing moment is defined as time 2). Subsequently, the 3-D hyperspectral tensor subspace reconstruction network is employed along with the time 2 hyperspectral tensor subspace features with low spatial resolution and the time 2 MSI to reconstruct the time 2 HSI with high spatial resolution. In the experiment, we utilize three simulated datasets and two real datasets to evaluate the fusion performance of our proposed method. The results demonstrate that our method achieves high-quality fusion results, exhibits comparable performance, and has robustness and practicality.**

***Index Terms*— Hyperspectral, multispectral, spatial–temporal–spectral fusion, tensor subspaces.**

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Weiwei Sun and Gang Yang are with the Department of Geography and Spatial Information Techniques, Ningbo University, Ningbo 315211, China (e-mail: sunweiwei@nbu.edu.cn; yanggang@nbu.edu.cn).

Kai Ren and Xiangchao Meng are with the Faculty of Electrical Engineering and Computer Science, Ningbo University, Ningbo 315211, China (e-mail: kairen0908@outlook.com; mengxiangchao@nbu.edu.cn).

Jiangtao Peng is with the Hubei Key Laboratory of Applied Mathematics, Faculty of Mathematics and Statistics, Hubei University, Wuhan 430062, China (e-mail: pengjt1982@hubu.edu.cn).

Jiancheng Li is with the School of Geodesy and Geomatics, Wuhan University, Wuhan 430079, China, and also with the Key Laboratory of Geospace Environment and Geodesy, Ministry of Education and Joint International Research Laboratory of Modern Geodesy and Geodynamics, Ministry of Education, Wuhan University, Wuhan 430079, China (e-mail: jcli@whu.edu.cn).

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I. Introduction

SATELLITES are widely utilized in various fields, such as environmental monitoring, agricultural monitoring, and resource exploration due to their fine spectral resolution [1], [2]. However, these satellites suffer from limitations in spatial resolution due to their narrow wavelength range, resulting in limited absorption of reflected energy [3]. In addition, hyperspectral satellites have long revisit periods, and the quality of the acquired data is susceptible to cloud cover and noise, leading to reduced effective data acquisition frequency and low temporal resolution, which greatly hampers the practical application of hyperspectral images (HSIs) [4], [5]. In contrast, multispectral satellite data have high spatial resolution and shorter revisit periods [6]. This enables a higher frequency of effective data acquisition and improves temporal resolution [7]. However, multispectral images (MSIs) possess coarse spectral resolution and are unsuitable for fine surface monitoring [8]. Consequently, spatial–temporal–spectral fusion of HSI and MSI aims to enhance the temporal and spatial resolution of HSI by leveraging MSI as auxiliary information [9].

Currently, image fusion methods are mainly divided into spatial–spectral fusion methods (SSF) [10], spatial–temporal fusion methods (STF) [11], and spatial–temporal–spectral fusion methods (STSF) [12].

SSF methods aim to enhance the spatial resolution of HSI by fusing the spectral information of HSI with the spatial information of MSI acquired at the same time, without considering the improvement of temporal resolution [13]. SSF methods encompass various approaches such as component substitution (CS), multiresolution analysis (MRA), iterative optimization, and deep learning methods [14]. CS methods assume that MSI subspace can represent the HSI subspace. Forward transformation is performed to obtain the HSI subspace, and then, inverse transformation is conducted using MSI to replace the HSI subspace, thus improving the spatial resolution of HSI [15]. Typical methods include intensity–hue–saturation (IHS)

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transformation [16]. Principal component analysis (PCA) for SSF has been proposed by Pal et al. [17], and adaptive Gram–Schmidt (GSA) has been introduced by Aiazzi et al. [18]. Although CS methods directly incorporate spatial details from MSI into the fused image, they alter the spectral characteristics of the original HSI, leading to serious spectral distortion in the fused image [7]. MRA methods assume that the information difference between images exists at multiscale features. By calculating the difference information between images at different scales through multiscale decomposition, SSF is achieved by injecting the difference information into HSI [19]. Typical approaches include modulation transfer function-generalized Laplacian pyramid (MTF-GLP) [20] and "a trous" wavelet transform (ATWT) [21]. MRA methods maintain better spectral fidelity as they do not change the spectral characteristics of HSI. However, some spatial information from MSI is not fully integrated into the fused image, leading to limited spatial resolution enhancement [9]. Iterative optimization methods usually decompose HSI and MSI into spectral matrix and spatial matrix and fuse the images by recombining the spectral matrix of HSI with the spatial matrix of MSI [14]. Yokoya et al. [15] incorporated image unmixing theory into SSF by recombining the HSI endmember matrix and MSI abundance matrix. Nezhad et al. [22] proposed the nonlocal sparse representation method based on pixel group using sparse representation theory. Dian et al. [23] presented nonlocal sparse tensor factorization (NLSTF). Iterative optimization methods typically model HSI and MSI based on matrix decomposition and solve them through methods such as alternating direction method of multipliers (ADMM) or augmented Lagrangian method (ALM), which yield comparable fusion results. However, these methods often involve numerous parameters, exhibit poor robustness, and have low computational efficiency [7]. Deep learning methods extract deep expressive features from HSI and MSI and employ nonlinear regression and extensive datasets for network training to obtain a universal SSF model [9]. These methods can be categorized into supervised and unsupervised methods. Supervised methods require the calculation of real label loss for network training. For instance, Palsson et al. [24] utilized a convolutional neural network (CNN) for image fusion. Xiao et al. [25] designed an SSF network based on generative adversarial network (GAN), and Bandara and Patel [26] introduced the transformer network to enhance HSI spatial resolution. However, supervised methods face challenges in obtaining real labels for HSI datasets. Some studies have employed simulated datasets for network training and applied the trained network to real datasets for fusion. Nonetheless, due to the uncertainties in the simulation process, supervised methods often exhibit poor fusion performance on real datasets. On the other hand, unsupervised methods achieve image fusion by incorporating data degradation priors into the network for self-supervised training. Wang et al. [27] proposed an unsupervised convolutional variational network for hyperspectral and MSI fusion. Li et al. [28] developed a novel unsupervised HSI–MSI fusion network with adaptive learning for degradation, and Liu et al. [29] constructed an unsupervised multiattention-guided network (UMAG-Net) that does not rely on training data to achieve improved HSI–MSI fusion. Unsupervised methods do not require real labels and yield better fusion results and application performance.

STF methods aim to use high temporal resolution and low spatial resolution remote sensing images to predict high temporal and spatial resolution remote sensing images [30]. These methods primarily focus on the fusion of moderate-resolution imaging spectroradiometer (MODIS) and Landsat images, which effectively enhances the temporal resolution of Landsat data while not improving the spectral resolution of Landsat images [12]. STF methods can be categorized into weight function methods, unmixing methods, Bayesian methods, and deep learning methods [31]. Weight function methods achieve precise pixel values by linearly combining the input image information [31]. Notable methods include the spatial and temporal adaptive reflectance fusion model (STARFM) proposed by Gao et al. [32], enhanced STARFM (ESTARFM) proposed by Zhu et al. [33], the improved weight function of STARFM introduced by Shen et al. [34], and the utilization of land cover data to enhance ESTARFM proposed by Fu et al. [35]. Weight function methods deliver accurate predictions for simple surfaces, but their performance is limited when dealing with complex underlying surfaces [36]. Unmixing methods assume that coarse pixels are mixed pixels and estimate accurate pixel values by unmixing the coarse pixels [37]. Typical methods include the spatial and temporary data fusion model (STDFM) proposed by Zhang et al. [38] and a locally corrected multiple expression model proposed by Moriondo et al. [39] for predicting the normalized difference vegetation index (NDVI) based on Landsat and MODIS images. Similar to weight function methods, these approaches are not effective in predicting images with complex underlying surfaces [37]. Bayesian methods transform the STF problem into a probability distribution problem and obtain the fused image by maximizing the posterior probability density [36]. Representative methods include the use of different linear models by Huang et al. [40] to establish temporal and spatial relationships between Landsat and MODIS images and the application of Bayesian maximum entropy by Li et al. [41] for pixel prediction. However, Bayesian methods still struggle to achieve satisfactory results for images with complex underlying surfaces [37]. With the advancement of deep learning, CNN, GAN, and transformer networks have been employed to predict accurate Landsat images based on MODIS images. Song et al. [42] proposed a spatiotemporal fusion method based on deep CNN for massive remote sensing data, which yielded satisfactory results for Landsat and MODIS data. Zhang et al. [43] introduced a GAN-based remote sensing image spatiotemporal fusion method (STFGAN). STFGAN significantly improved prediction accuracy by utilizing residual blocks and two pairs of Landsat and MODIS images. Chen et al. [44] proposed a novel algorithm based on the Swin transformer and linear unmixing theory for Landsat and MODIS image fusion. Deep learning methods achieve high-quality prediction of Landsat data; however, they require a large number of training samples [12].

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The objective of STSF methods is to integrate temporal, spatial, and spectral information from multiple sources to reconstruct remote sensing images with high temporal, spatial, and spectral resolution [12]. These methods can be categorized into step-by-step fusion methods and integrated fusion methods [45]. Step-by-step fusion methods divide the STSF process into multiple stages, such as combining STF and SSF to achieve STSF [46]. For instance, Zhang et al. [47] proposed a SPAtial Temporary Spectral blending model (SPATS) for predicting surface reflectivity with high temporal, spatial, and spectral resolution. SPATS fused Landsat, Hyperion, and MODIS data based on existing STF and SSF models. Zhao and Huang [48] developed a hybrid spatial–spectral image fusion model (HSTSFM) that integrates temporal, spatial, and spectral information from Landsat, Hyperion, and MODIS data. HSTSFM consists of three fusion modules: SSF, STF, and temporal–spectral fusion. Step-by-step fusion methods are relatively straightforward; however, they may suffer from error accumulation between models, leading to severe image distortion [12]. Integrated fusion methods model the input data uniformly to obtain remote sensing images with high temporal, spatial, and spectral resolution [40]. Shen et al. [11] proposed an integrated STSF framework for remote sensing images, which does not impose limitations on the number of remote sensing sensors. Peng et al. [49] introduced an integrated spatiotemporal–spectral fusion framework based on microscopic sparse tensor factorization and applied it to STSF using simulated datasets. Integrated fusion methods effectively mitigate the issue of error accumulation. However, these models often involve numerous parameters, exhibit poor robustness, and have low computational efficiency [12].

In general, traditional SSF methods can enhance the spatial and spectral resolution of remote sensing images but do not improve the temporal resolution. On the other hand, STF methods can improve the temporal and spatial resolution but cannot enhance the spectral resolution. Existing STSF methods can improve the temporal, spatial, and spectral resolution of remote sensing images, but they typically require at least three types of remote sensing images (low spatial and temporal resolution with high spectral resolution, low temporal and spectral resolution with high spatial resolution, and low spatial and spectral resolution with high temporal resolution). However, collecting different types of remote sensing data simultaneously is challenging due to sensor design and weather conditions. Although some studies have attempted to fuse HSI with low temporal and spatial resolution and MSI with high temporal and spatial resolution to obtain HSI with high temporal and spatial resolution (requiring only two types of remote sensing data), such studies are rare. Furthermore, to the best of our knowledge, only Zhou et al. [12] proposed a generalized linear spectral mixing model for HSI and MSI (GLMM-STSF). However, GLMM-STSF has numerous constraints and parameters, making it prone to convergence issues and exhibiting low robustness, low computational efficiency, and limited fusion performance.

Extensive research has shown the nonlinear relationship between HSI and MSI, and deep learning methods have demonstrated comparable fusion performance to traditional linear methods [8], [26], [50]. To enhance the practicality and robustness of STSF and obtain high-quality fused data, this article proposes an unsupervised 3-D tensor subspace decomposition network for spatial–temporal–spectral fusion of HSI and MSI (TSDN). TSDN aims to reduce the modality difference between HSI and MSI. It first constructs a 3-D HSI/MSI tensor subspace extraction network to generate upsampled T1 HSI tensor subspaces and Gaussian blurred T1 MSI tensor subspaces. The loss between tensor subspaces is calculated to approximate T1 upsampled HSI and Gaussian blurred MSI pairs, aligning them in terms of modal features. Subsequently, the 3-D MSI tensor subspace extraction network trained on Gaussian blurred T1 MSI is applied to Gaussian blurred T2 MSI, obtaining tensor subspaces for Gaussian blurred T2 MSI. The difference features are obtained by subtracting the Gaussian blurred T1 MSI tensor subspaces from the Gaussian blurred T2 MSI tensor subspaces. The T2 HSI tensor subspaces with low spatial resolution are obtained by adding the difference features to the upsampled T1 HSI tensor subspaces. TSDN then constructs a 3-D HSI tensor subspace reconstruction network (TSN) using T2 HSI tensor subspaces with low spatial resolution and T2 MSI as input data to reconstruct T2 HSI with high spatial resolution. Finally, TSDN takes the reconstructed T2 HSI as input data and employs the trained 3-D HSI tensor subspace extraction network to generate T2 HSI tensor subspaces. The 3-D HSI TSN and 3-D HSI tensor subspace extraction network are continuously optimized by calculating the loss between T2 HSI tensor subspaces and T2 MSI tensor subspaces. In addition, to preserve the band relationship of the fused image and enhance spectral fidelity, TSDN employs the trained 3-D HSI TSN and T1 HSI tensor subspaces to reconstruct the upsampled T1 HSI. The loss between the reconstructed T1 HSI and the original T1 HSI further optimizes the 3-D HSI TSN. The main contributions of this article are given as follows.

1) The proposed TSDN utilizes deep learning to approximate tensor subspace decomposition theory, enhancing interpretability, providing a nonlinear solution, and improving convergence speed. It effectively avoids the computational burden associated with alternative direction method of multipliers (ADMM) and ALM while achieving satisfactory fusion results with fewer parameters.
2) TSDN constructs a 3-D HSI tensor subspace extraction network, a 3-D MSI tensor subspace extraction network, and a 3-D HSI TSN. A reasonable loss function is designed to realize unsupervised STSF of HSI and MSI, significantly enhancing the fidelity of fused images.
3) TSDN demonstrates comparative fusion performance in both simulated and real datasets, outperforming five state-of-the-art methods in terms of temporal prediction, spatial resolution, and spectral fidelity.

The structure of this article is organized as follows. Section II provides a detailed introduction to the proposed TSDN. Section III presents the experimental results and analysis. Finally, Section III-E concludes this article.

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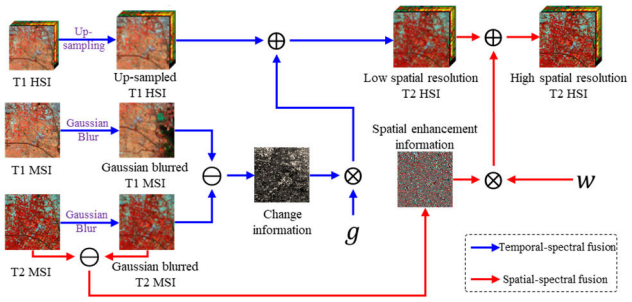


Fig. 1. Generalized spatial–temporal–spectral fusion framework.

II. Methodology

In this section, an overview of our proposed TSDN is first presented to show the whole process of STSF. Then, the structure of TSDN is explained concretely. Finally, we describe the design of the loss function.

A. Overview of TSDN

TSDN uses HSI observed at T1 and MSI at T1 and MSI at T2 to predict HSI at T2. Specifically, T2 HSI with high spatial resolution can be obtained by injecting change information between T2 MSI and T1MSI and the spatial information between T2 MSI and Gaussian blurred T2 MSI to T1 HSI, and this generalized formula can be defined as follows:

$$\mathbf{H}_{t2} = \tilde{\mathbf{H}}_{t1} + g(\bar{M}_{t2} - \bar{\mathbf{M}}_{t1}) + w(\mathbf{M}_{t2} - \bar{\mathbf{M}}_{t2}) \tag{1}$$

where $\mathbf{H}_{t2} \in \mathbb{R}^{a\times b\times K}$ represents T2 HSI with high spatial resolution; $\tilde{\mathbf{H}}_{t1} \in \mathbb{R}^{A\times B\times K}$ represents upsampled T1 HSI; $\mathbf{M}_{t2} \in \mathbb{R}^{A\times B\times k}$ is the T2 MSI; $\bar{\mathbf{M}}_{t1} \in \mathbb{R}^{A\times B\times k}$ and $\bar{\mathbf{M}}_{t2} \in \mathbb{R}^{A\times B\times k}$ represent Gaussian blurred T1 and T2 MSI, respectively; a, b, and K are the number of rows, columns, and bands of $\mathbf{H}$, respectively; and A, B, and k are the number of rows, columns, and bands of $\mathbf{M}$, respectively, $a < A, b < B$, and $k < K$. g is the change gain matrix, and w is the spatial gain matrix. Fig. 1 shows our proposed spatial–temporal–spectral fusion framework. It is worth noting that in the experiment, we use the bicubic algorithm to interpolate the T1 HSI to obtain the T1 HSI being upsampled. In addition, a 5×5 Gaussian kernel used to blur the T1 and T2 MSI to obtain Gaussian blurred T1 and T2 MSI. However, due to the serious difference of spatial–temporal–spectral information between HSI and MSI, it is difficult to obtain g and w by using the traditional linear model.

To model TSDN for change information prediction, we introduce the theory of subspace extraction and tensor decomposition [51]. The reason is that, on the one hand, the introduction of subspace can achieve modal unity for HSI and MSI, effectively avoiding the mismatch between the change information and the T1HS data characteristics; on the other hand, the change information between images at different times exists in row-x, column-y, and band-z, and the introduction of tensor decomposition theory effectively improves the change prediction of fused images.

First, HSI and MSI observed at the same time have spectral and spatial differences due to different resolutions. HSI can be transformed into MSI by spectral response function and spatial resolution enhancement, and MSI can also be transformed into HSI by spectral resolution enhancement and point distribution function. Therefore, to unify the modal between data, we assume that T1 HSI and MSI have a common subspace, which can be expressed as follows:

$$\mathbf{S}_1 = \Psi_1\left(\tilde{\mathbf{H}}_{t1}\right) \;\&\; \mathbf{S}_2 = \Psi_2\left(\bar{\mathbf{M}}_{t1}\right), \quad \text{s.t. } \mathbf{S}_1 \approx \mathbf{S}_2 \tag{2}$$

where Ψ_1 and Ψ_2 are the subspace extraction equations of $\tilde{\mathbf{H}}_{t1}$ and $\bar{\mathbf{M}}_{t1}$, respectively (that is subspace extraction network in TSDN), $\mathbf{S}_1 \in \mathbb{R}^{A\times B\times\beta}$ and $\mathbf{S}_2 \in \mathbb{R}^{A\times B\times\beta}$ are the subspace of $\tilde{\mathbf{H}}_{t1}$ and $\bar{\mathbf{M}}_{t1}$, respectively, and β is the number of subspaces.

Meanwhile, according to the tensor decomposition theory, the observed HSI and MSI are regarded as a 3-D tensor, that is, the image contains x-spatial dimension, y-spatial dimension, and z-spectral dimension, which can be decomposed into three tensor matrices of different dimensions, and the original image can be reconstructed through the tensor matrices [52]. The mathematical expression model for tensor decomposition is given as follows:

$$\mathbf{I} = \mathcal{G} \times \mathbf{x}_{\text{tensor}} \times \mathbf{y}_{\text{tensor}} \times \mathbf{z}_{\text{tensor}} \tag{3}$$

where $\mathbf{I} = [\tilde{\mathbf{H}}_{t1}, \bar{\mathbf{M}}_{t1}]$ is the image. $\mathbf{x}_{\text{tensor}} \in \mathbb{R}^{m\times p}$, $\mathbf{y}_{\text{tensor}} \in \mathbb{R}^{n\times q}$, and $\mathbf{z}_{\text{tensor}} \in \mathbb{R}^{l\times r}$ are factor matrices with different dimensions, and p, q, and r are column vector of $\mathbf{x}_{\text{tensor}}$, $\mathbf{y}_{\text{tensor}}$, and $\mathbf{z}_{\text{tensor}}$, respectively. $\mathcal{G} \in \mathbb{R}^{p\times q\times r}$ is a kernel tensor. However, traditional mathematical expression models cannot perform nonlinear operations in deep learning models. Therefore, we used convolution operation to approximate the tensor decomposition process in TSDN, namely

$$\left[\mathbf{x}_{\text{tensor}}, \mathbf{y}_{\text{tensor}}, \mathbf{z}_{\text{tensor}}\right] = f_t(\mathbf{I}) \tag{4}$$

$$\mathbf{I} = c(\mathbf{x}_{\text{tensor}}, \mathbf{y}_{\text{tensor}}, \mathbf{z}_{\text{tensor}}) \tag{5}$$

where f_t represents the tensor decomposition network, $\mathbf{x}_{\text{tensor}}$, $\mathbf{y}_{\text{tensor}}$, and $\mathbf{z}_{\text{tensor}}$ represent tensor matrices in the x-, y-, and z-directions by convolution operation, respectively, and unlike (3), $\mathbf{x}_{\text{tensor}}$, $\mathbf{y}_{\text{tensor}}$, and $\mathbf{z}_{\text{tensor}}$ are matrices of $A \times B \times \beta$. c is the reconstruction network, and the network parameters can be approximated as kernel tensors. Therefore, by combining (2), (4), and (5), we decompose the upsampled HSI and Gaussian blurred MSI observed at T1 into tensor subspaces of different dimensions. The formula is redefined as follows:

$$\begin{aligned}
&\left[\mathbf{x}_{ts}^{T1H}, \mathbf{y}_{ts}^{T1H}, \mathbf{z}_{ts}^{T1H}\right] = \Psi_H\left(f_t^h\left(\tilde{\mathbf{H}}_{t1}\right)\right) \\
&\left[\mathbf{x}_{ts}^{T1M}, \mathbf{y}_{ts}^{T1M}, \mathbf{z}_{ts}^{T1M}\right] = \Psi_M\left(f_t^m\left(\bar{\mathbf{M}}_{t1}\right)\right) \\
&\text{s.t. } \mathbf{x}_{ts}^{T1H} \approx \mathbf{x}_{ts}^{T1M}, \quad \mathbf{y}_{ts}^{T1H} \approx \mathbf{y}_{ts}^{T1M}, \quad \mathbf{z}_{ts}^{T1H} \approx \mathbf{z}_{ts}^{T1M}
\end{aligned} \tag{6}$$

where $\mathbf{x}_{ts}^{T1H} \in \mathbb{R}^{A\times B\times\beta}$, $\mathbf{y}_{ts}^{T1H} \in \mathbb{R}^{A\times B\times\beta}$, and $\mathbf{z}_{ts}^{T1H} \in \mathbb{R}^{A\times B\times\beta}$ are tensor subspace matrices of $\tilde{\mathbf{H}}_{t1}$7. $\mathbf{x}_{ts}^{T1M} \in \mathbb{R}^{A\times B\times\beta}$, $\mathbf{y}_{ts}^{T1M} \in \mathbb{R}^{A\times B\times\beta}$, and $\mathbf{z}_{ts}^{T1M} \in \mathbb{R}^{A\times B\times\beta}$ are the tensor subspace matrices of $\bar{\mathbf{M}}_{t1}$; Ψ_H and Ψ_M are the tensor subspace extraction equations of HSI and MSI, respectively; and f_t^h and f_t^m are the tensor decomposition equations of $\tilde{\mathbf{H}}_{t1}$ and $\bar{\mathbf{M}}_{t1}$, respectively.

In addition, there are often spatial and spectral differences between MSI observed at different times. In other words, the change between images occurs in the subspaces of x, y, and z. We assume that the tensor subspace extraction of MSI

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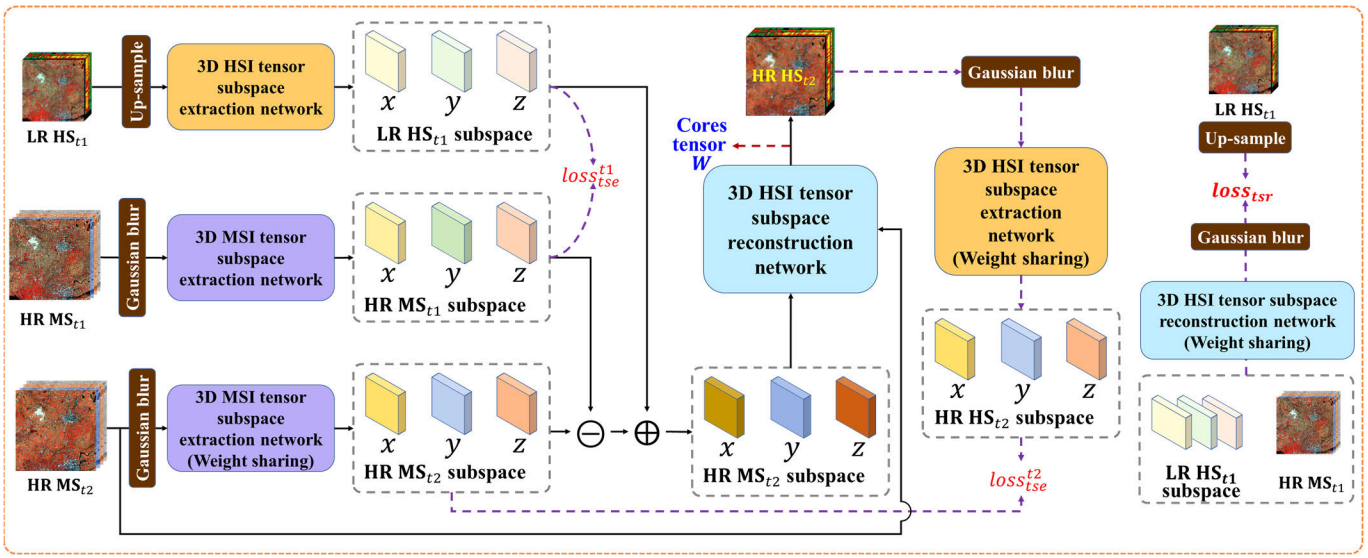


Fig. 2. TSDN.

at different times follows the same rule, that is, there is a universal Ψ_M and f_t^m. Therefore, the T2 MSI can be expressed as follows:

$$\left[\mathbf{x}_{ts}^{T2M}, \mathbf{y}_{ts}^{T2M}, \mathbf{z}_{ts}^{T2M}\right] = \Psi_M\left(f_t^m\left(\bar{\mathbf{M}}_{t2}\right)\right) \tag{7}$$

where $\mathbf{x}_{ts}^{T2M} \in \mathbb{R}^{A\times B\times\beta}$, $\mathbf{y}_{ts}^{T2M} \in \mathbb{R}^{A\times B\times\beta}$, and $\mathbf{z}_{ts}^{T2M} \in \mathbb{R}^{A\times B\times\beta}$ represent the tensor subspaces of $\bar{\mathbf{M}}_{t2}$.

Then, the tensor subspace matrices of the predicted T2 HSI with low spatial resolution can be obtained by adding the difference information between tensor subspaces of Gaussian blurred MSI at different time to the tensor subspace of upsampled T1 HSI

$$\mathbf{x}_{ts}^{T2H} = \mathbf{x}_{ts}^{T1H} + \left(\mathbf{x}_{ts}^{T2M} - \mathbf{x}_{ts}^{T1M}\right) \tag{8}$$

$$\mathbf{y}_{ts}^{T2H} = \mathbf{y}_{ts}^{T1H} + \left(\mathbf{y}_{ts}^{T2M} - \mathbf{y}_{ts}^{T1M}\right) \tag{9}$$

$$\mathbf{z}_{ts}^{T2H} = \mathbf{z}_{ts}^{T1H} + \left(\mathbf{z}_{ts}^{T2M} - \mathbf{z}_{ts}^{T1M}\right). \tag{10}$$

To model TSDN for spatial information prediction, the fused T2 HSI with high spatial resolution can be eventually obtained by using $\mathbf{M}_{t2}$ as an auxiliary feature with high spatial resolution

$$\mathbf{H}_{t2} = c\left(\mathbf{x}_{ts}^{T2H}, \mathbf{y}_{ts}^{T2H}, \mathbf{z}_{ts}^{T2H}, \mathbf{M}_{t2}\right). \tag{11}$$

For the above formula, the unknown variables include f_t^h, f_t^m, Ψ_H, and Ψ_M, c. Since f_t^h and Ψ_H jointly extract the tensor subspace matrices of HSI and f_t^m and Ψ_M jointly extract the tensor subspace matrices of MSI, $[f_t^h, \Psi_H]$ can be redefined as g_{ts}^h, and $[f_t^m, \Psi_M]$ can be redefined as g_{ts}^m. Therefore, TSDN first establishes the 3-D HSI tensor subspace extraction network g_{ts}^h and the 3-D MSI tensor subspace extraction network g_{ts}^m to obtain the tensor subspace matrices of $\tilde{\mathbf{H}}_{t1}$ and $\bar{\mathbf{M}}_{t1}$. Then, the 3-D MSI tensor subspace extraction network trained at $\bar{\mathbf{M}}_{t1}$ is mapped to $\bar{\mathbf{M}}_{t2}$ to obtain the tensor subspace matrices of $\bar{\mathbf{M}}_{t2}$. After that, the tensor subspace matrices of $\bar{\mathbf{M}}_{t2}$ are subtracted from the tensor subspace matrices of $\bar{\mathbf{M}}_{t1}$ to obtain the difference matrices, and the difference matrices are added to the tensor subspace matrices of $\tilde{\mathbf{H}}_{t1}$ to obtain the tensor subspace matrices of T2 HSI with low spatial resolution. Finally, we established the 3-D HSI TSN, and the obtained tensor subspace matrices of $\tilde{\mathbf{H}}_{t2}$ and $\mathbf{M}_{t2}$ are used as input data to obtain $\mathbf{H}_{t2}$. It is worth noting that in the experiment, the network parameter of TSN is finally regarded as c. TSDN is shown in Fig. 2.

There is, in fact, no explicit one-to-one match from (2) to (10) to the canonical mathematical representation of subspace extraction and tensor decomposition. Yet, we argue that TSDN can adaptively learn the matrix factorization and reconstruction process in a data-driven manner.

B. Network Architecture

TSDN uses three network modules for joint training, including 3-D HSI tensor subspace extraction network (g_{ts}^h), 3-D MSI tensor subspace extraction network (g_{ts}^m), and 3-D HSI TSN.

1) 3-D HSI Tensor Subspace Extraction Network (g_{ts}^h): Tensor subspace decomposition is widely used in image reconstruction. Therefore, many SSF and STF methods based on tensor subspace decomposition have been proposed [53], [54], [55]. However, these methods for STSF need to be optimized by ALM and ADMM, which are computationally complex. We propose a 3-D HSI tensor subspace extraction network (g_{ts}^h), which can effectively simulate the process of tensor subspace decomposition and is computationally efficient. To make it easy to understand, we externalize the g_{ts}^h process in Fig. 3. However, 3-D convolutional layers are notorious for their high computing and storage costs. Therefore, 2-D convolution (f_2) is used in our work to approximate 3-D tensor subspace decomposition. To decompose the spatial x-tensor, the T1 HSI is used as input data, the 2-D convolution kernel of 3×1 is used, the step size is 1, and the pooling is (1, 0). $\mathbf{x}_{ts}^{T1H}$ is obtained after five layers of convolution. This process can be

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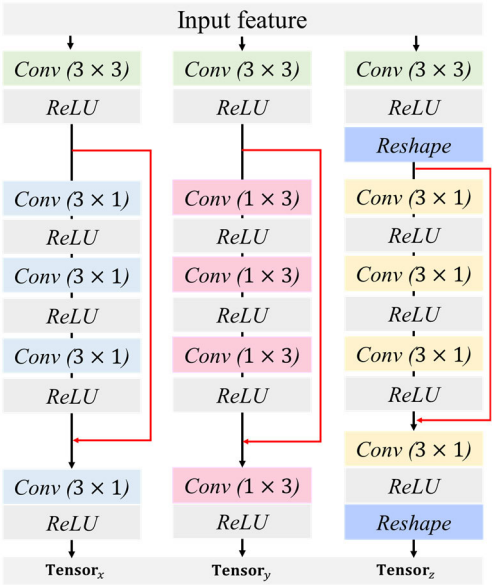


Fig. 3. 3-D HSI tensor subspace extraction network (g_{ts}^{h}).

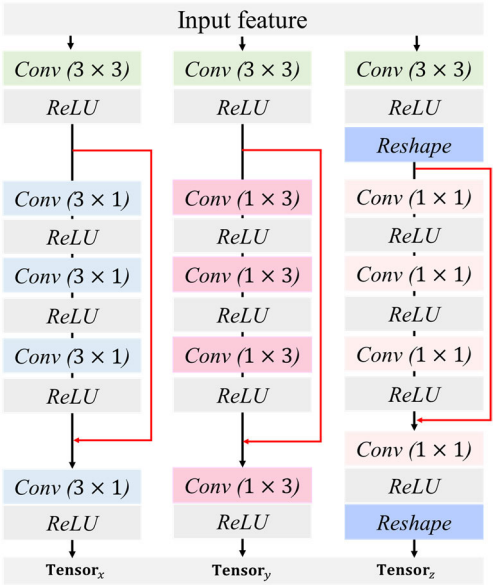


Fig. 4. 3-D MSI tensor subspace extraction network (g_{ts}^{m}).

defined as follows:

$$\mathbf{x}_{ts}^{T1H} = \mathrm{Relu}\big(f_2^{3\times1}\big(\mathrm{Relu}\big(f_2^{3\times1}(\mathrm{Relu} \times \big(f_2^{3\times1}\big(\mathrm{Relu}\big(f_2^{3\times1}\big(\mathrm{Relu}\big(f_2^{3\times1}(\tilde{\mathbf{H}}_{t1})\big)\big)\big)\big)\big)\big)\big)\big) \quad (12)$$

where $f_2^{3\times1}$ is a 2-D convolution operator with 3×1 convolution kernel and Relu is the activation function.

To decompose the spatial-y tensor, the T1 HSI is used as input data, the 2-D convolution kernel of 1×3 is used, the step size is 1, and the pooling is (0, 1). After five layers of convolution, $\mathbf{y}_{ts}^{T1H}$ is obtained. This process can be defined as follows:

$$\mathbf{y}_{ts}^{T1H} = \mathrm{Relu}\big(f_2^{1\times3}\big(\mathrm{Relu}\big(f_2^{1\times3}(\mathrm{Relu} \times \big(f_2^{1\times3}\big(\mathrm{Relu}\big(f_2^{1\times3}\big(\mathrm{Relu}\big(f_2^{1\times3}(\tilde{\mathbf{H}}_{t1})\big)\big)\big)\big)\big)\big)\big)\big) \quad (13)$$

where $f_2^{1\times3}$ is the 2-D convolution operator of the 1×3 convolution kernel.

Considering that 2-D convolution cannot be directly applied to decompose $\mathbf{z}_{ts}^{T1H}$, we first performed the reshape operation on HSI, which transforms $A \times B \times K$ of the original data into $K \times A \times B$. Then, the 2-D convolution kernel of 3×1 is used, the step is 1, and the pooling is (1, 0). After five convolution layers, the reshape is used to convert $K \times A \times B$ to $A \times B \times K$ to obtain the final $\mathbf{z}_{ts}^{T1H}$, and this process can be defined as follows:

$$\mathbf{z}_{ts}^{T1H} = R^{A\times B\times K}\big(\mathrm{Relu}\big(f_2^{3\times1}\big(\mathrm{Relu}\big(f_2^{3\times1}\big(\mathrm{Relu}\big(f_2^{3\times1} \times \big(\mathrm{Relu}\big(f_2^{3\times1}\big(\mathrm{Relu}\big(f_2^{3\times1}\big(R^{z\times x\times y}(\tilde{\mathbf{H}}_{t1})\big)\big)\big)\big)\big)\big)\big)\big)\big)\big) \quad (14)$$

where $R^{A\times B\times K}$ and $R^{K\times A\times B}$ are reshape operations.

It is worth noting that in order to ensure that HSI and MSI have a unified modal, g_{ts}^{h} converts $\mathbf{x}_{ts}^{T1H}$, $\mathbf{y}_{ts}^{T1H}$, and $\mathbf{z}_{ts}^{T1H}$ into 64 features.

2) 3-D MSI Tensor Subspace Extraction Network (g_{ts}^{m}): The design of MSI tensor subspace extraction network (g_{ts}^{m}) is almost consistent with g_{ts}^{h}. However, due to the small number of bands in MSI, we use a 1×1 convolution kernel for z (see Fig. 4).

g_{ts}^{m} also uses the 2-D convolution kernel of 3×1 when obtaining $\mathbf{x}_{ts}^{T1M}$, the step size is 1, the pooling is (1, 0), and the number of convolution layers is 5. Five convolution layers were also used to obtain $\mathbf{y}_{ts}^{T1M}$, the convolution kernel is 1×3, the step size is 1, and the pooling is (0, 1). This process is defined as follows:

$$\mathbf{x}_{ts}^{T1M} = \mathrm{Relu}\big(f_2^{3\times1}\big(\mathrm{Relu}\big(f_2^{3\times1}(\mathrm{Relu} \times \big(f_2^{3\times1}\big(\mathrm{Relu}\big(f_2^{3\times1}\big(\mathrm{Relu}\big(f_2^{3\times1}(\bar{\mathbf{M}}_{t1})\big)\big)\big)\big)\big)\big)\big)\big) \quad (15)$$

$$\mathbf{y}_{ts}^{T1M} = \mathrm{Relu}\big(f_2^{1\times3}\big(\mathrm{Relu}\big(f_2^{1\times3}(\mathrm{Relu} \times \big(f_2^{1\times3}\big(\mathrm{Relu}\big(f_2^{1\times3}\big(\mathrm{Relu}\big(f_2^{1\times3}(\bar{\mathbf{M}}_{t1})\big)\big)\big)\big)\big)\big)\big)\big). \quad (16)$$

In addition, g_{ts}^{m} and g_{ts}^{h} have similar operations when decomposing $\mathbf{z}_{ts}^{T1M}$. A process is defined as follows:

$$\mathbf{z}_{ts}^{T1M} = R^{A\times B\times K}\big(\mathrm{Relu}\big(f_2^{1\times1}\big(\mathrm{Relu}\big(f_2^{1\times1}\big(\mathrm{Relu}\big(f_2^{1\times1} \times \big(\mathrm{Relu}\big(f_2^{1\times1}\big(\mathrm{Relu}\big(f_2^{1\times1}\big(R^{K\times A\times B}(\bar{\mathbf{M}}_{t1})\big)\big)\big)\big)\big)\big)\big)\big)\big)\big) \quad (17)$$

where $f_2^{1\times1}$ is the 2-D convolution operator of the 1×1 convolution kernel.

Specifically, the number of features in g_{ts}^{m} increases gradually, and the final number of features in the subspaces is 64.

3) 3-D HSI TSN: The reconstruction of H_{t2} is an important step by using tensor subspace (see Fig. 5). TSN first performs a 3×3 convolution for $\mathbf{x}_{ts}^{T2H}$, $\mathbf{y}_{ts}^{T2H}$, and $\mathbf{z}_{ts}^{T2H}$. Since different feature layers contain different information, these feature layers are superimposed. A process is defined as follows:

$$\mathbf{feature} = \mathrm{concat}\big(\mathrm{Relu}\big(f_2^{3\times3}\big(\mathbf{x}_{ts}^{T2H}\big), \mathrm{Relu}\big(f_2^{3\times3}\big(\mathbf{y}_{ts}^{T2H}\big), \mathrm{Relu}\big(f_2^{3\times3}\big(\mathbf{z}_{ts}^{T2H}\big), 3\big) \quad (18)$$

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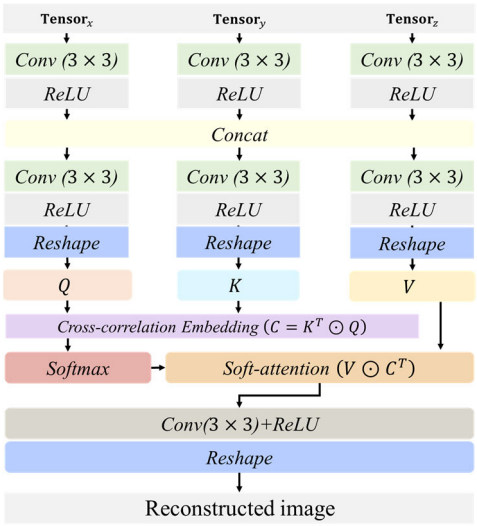


Fig. 5. 3-D HSI TSN.

where **feature** is the feature layer obtained after the convolution operation, $f_2^{3\times3}$ is the 2-D convolution operator with 3×3 convolution kernel, and concat$(\cdot, 3)$ is the feature concatenated on the third dimension.

Then, three convolution layers are used to operate on $feature$ to obtain feature images with different channels, and we reshaped these features into 2-D images. The specific formula is given as follows:

$$\mathbf{Q} = R^{AB\times64}\left(\text{Relu}\left(f_2^{3\times3}(\textbf{feature})\right)\right) \tag{19}$$

$$\mathbf{K} = R^{AB\times K}\left(\text{Relu}\left(f_2^{3\times3}(\textbf{feature})\right)\right) \tag{20}$$

$$\mathbf{V} = R^{AB\times64}\left(\text{Relu}\left(f_2^{3\times3}(\textbf{feature})\right)\right). \tag{21}$$

After that, $\mathbf{Q}$ and $\mathbf{K}^T$ are multiplied to get the matrix $\mathbf{C}$, and T is the transpose operation. Considering the unbounded convolution operation, we use softmax to normalize the matrix $\mathbf{C}$. The final reconstructed image is obtained by multiplying the matrix $\mathbf{V}$ and $\mathbf{C}^T$, a reshaping operation, and a 3×3 convolution layer

$$\mathbf{C} = \text{softmax}(\mathbf{K}^T \odot \mathbf{Q}) \tag{22}$$

$$\overline{\overline{\mathbf{H}}}_{t2} = R^{A\times B\times K}\left(\text{Relu}\left(f_2^{3\times3}\left(\mathbf{V} \odot \mathbf{C}^T\right)\right)\right) \tag{23}$$

where $\overline{\overline{\mathbf{H}}}_{t2}$ is T2 HSI with low spatial resolution and $\odot$ is the dot product.

However, $\mathbf{x}_{ts}^{T2H}$, $\mathbf{y}_{ts}^{T2H}$, and $\mathbf{z}_{ts}^{T2H}$ are derived from degraded images and have limited spatial resolution. To improve the spatial resolution of $\overline{\overline{\mathbf{H}}}_{t2}$, T2 MSI is used as an auxiliary feature with high spatial resolution to reconstruct T2 HSI. The specific formula is defined as follows:

$$\mathbf{H}_{t2} = \text{Relu}\left(f_2^{3\times3}\left(\text{Concat}(\overline{\overline{\mathbf{H}}}_{t2}, \mathbf{M}_{t2}, 3)\right)\right). \tag{24}$$

It is worth noting that in the experiment, we approximate the tensor reconstruction process by using the parameter f_θ^c obtained by TSN training as the kernel tensor. In fact, the classical mathematical representation from subspace extraction to tensor decomposition and reconstruction has no explicit one-to-one matching.

C. Loss Setting

As shown in Fig. 1, there are some important loss functions that need to be considered carefully.

To extract the tensor subspaces of T1 HSI and T1 MSI, TSDN uses mean squared error (MSE) to calculate loss_x^{t1}, loss_y^{t1}, and loss_z^{t1} to optimize the 3-D HSI/MSI tensor subspace extraction network, which can be expressed as follows:

$$\text{loss}_{tse}^{t1} = \text{loss}_x^{t1} + \text{loss}_y^{t1} + \text{loss}_z^{t1} \tag{25}$$

$$\text{loss}_x^{t1} = \frac{1}{N}\sum_{i=1}^{N}\left(\mathbf{x}_{ts}^{T1H} - \mathbf{x}_{ts}^{T1M}\right)^2 \tag{26}$$

$$\text{loss}_y^{t1} = \frac{1}{N}\sum_{i=1}^{N}\left(\mathbf{y}_{ts}^{T1H} - \mathbf{y}_{ts}^{T1M}\right)^2 \tag{27}$$

$$\text{loss}_z^{t1} = \frac{1}{N}\sum_{i=1}^{N}\left(\mathbf{z}_{ts}^{T1H} - \mathbf{z}_{ts}^{T1M}\right)^2 \tag{28}$$

where N is the number of samples.

Due to the consistency of the relationship between HSI bands, loss_{tsr} is designed to optimize the 3-D HSI TSN and reduce the spectral distortion of the T2 fused HSI, that is, the satisfactory 3-D HSI TSN considered by TSDN can not only reconstruct T2 HSI but also reconstruct T1 HSI from the tensor subspaces of T1 HSI. loss_{tsr} is expressed as follows:

$$\text{loss}_{tsr} = \frac{1}{N}\sum_{i=1}^{N}\left(\mathcal{G}\left(f_\theta^c\left(\mathbf{x}_{ts}^{T2H}, \mathbf{y}_{ts}^{T2H}, \mathbf{z}_{ts}^{T2H}\right)\right) - \tilde{\mathbf{H}}_{t1}\right)^2 \tag{29}$$

where $\mathcal{G}$ is a Gaussian blur operation.

To further constrain the reconstruction loss of T2 HSI and optimize the 3-D HSI tensor subspace extraction network, loss_{tse}^{t2} was designed. TSDN first used the 3-D HSI tensor subspace extraction network to extract the tensor subspaces of fused and Gaussian blurred T2 HSI and believed that the extracted tensor subspaces should be consistent with the tensor subspaces of Gaussian blurred T2 MSI. Therefore, loss_{tse}^{t2} is designed and expressed as follows:

$$\text{loss}_{tse}^{t2} = \text{loss}_x^{t2} + \text{loss}_y^{t2} + \text{loss}_z^{t2} \tag{30}$$

$$\text{loss}_x^{t2} = \frac{1}{N}\sum_{i=1}^{N}\left(\mathbf{x}_{ts}^{T2H} - \mathbf{x}_{ts}^{T2M}\right)^2 \tag{31}$$

$$\text{loss}_y^{t2} = \frac{1}{N}\sum_{i=1}^{N}\left(\mathbf{y}_{ts}^{T2H} - \mathbf{y}_{ts}^{T2M}\right)^2 \tag{32}$$

$$\text{loss}_z^{t2} = \frac{1}{N}\sum_{i=1}^{N}\left(\mathbf{z}_{ts}^{T2H} - \mathbf{z}_{ts}^{T2M}\right)^2. \tag{33}$$

For TSDN, the final loss is defined as follows:

$$\text{loss}_{\text{overall}} = \text{loss}_{tse}^{t1} + \text{loss}_{tsr} + \text{loss}_{tse}^{t2}. \tag{34}$$

III. Experiments

A. Experimental Datasets

To evaluate the fusion performance of TSDN, we used three simulated datasets and two real datasets (see Table I). Each simulated dataset includes two HSI observed at different

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TABLE I
INFORMATION OF EXPERIMENTAL DATA

Datasets		Sensor	Spatial resolution (m)	Spectral resolution (nm)	Acquisition times (Years. Month. day)	
					Training data	Training data
Simulated datasets	Dataset-1	HS (ZY-1E)	120	400-2500	2020.12.19 (HS_{t1} & MS_{t1}) 2021.06.05 (MS_{t2})	2021.06.05 (HS_{t2})
		MS (SPOT-4)	30	400-900		
	Dataset-2	HS (ZY-1E)	120	400-2500	2020.12.19 (HS_{t1} & MS_{t1}) 2021.06.05 (MS_{t2})	2021.06.05 (HS_{t2})
		MS (SPOT-4)	30	400-900		
	Dataset-3	HS (ZY-1E)	120	400-2500	2020.01.13 (HS_{t1} & MS_{t1}) 2021.03.09 (MS_{t2})	2021.03.09 (HS_{t2})
		MS (SPOT-4)	30	400-900		
Real datasets	Dataset-1	HS (ZY-1E)	30	400-2500	2021.09.29 & 2021.09.29 (HS_{t1} & MS_{t1}) 2021.12.19 (MS_{t2})	2021.12.19 (HS_{t2})
		MS (ZY-1E)	10	400-950		
	Dataset-2	HS (ZY-1E)	30	400-2500	2021.01.03 & 2021.01.03 (HS_{t1} & MS_{t1}) 2021.05.17 (MS_{t2})	2021.05.17 (HS_{t2})
		MS (S2A)	10	400-900		

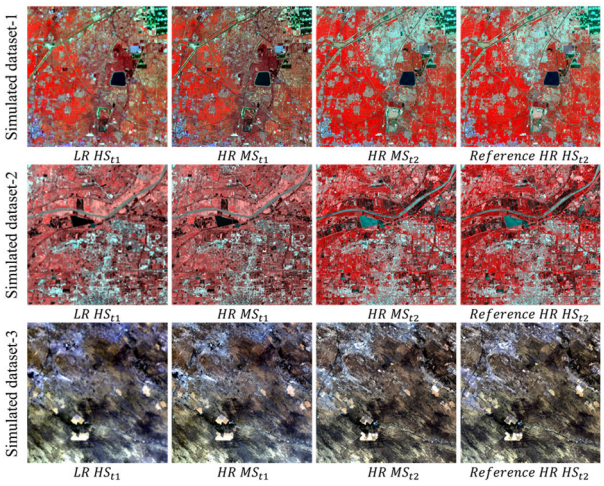


Fig. 6. Simulated datasets.

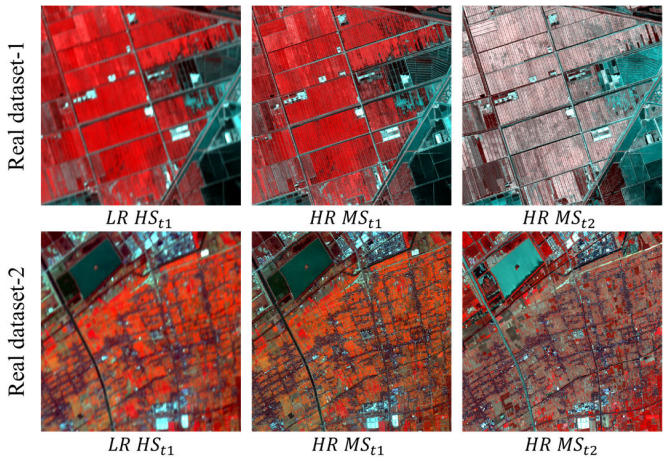


Fig. 7. Real datasets.

times. These HSI are from Zi Yuan (ZY)-1E hyperspectral satellite, and 166 bands with a spatial resolution of 30 m are obtained, with a spectral range of 400–2500 nm (ZY-1E HSI is downloaded from https://data.cresda.cn). In the experiment, we first preprocessed HSI, including ortho correction, radiometric calibration, and atmospheric correction, and finally eliminated the bands with serious water vapor absorption and noise. These preprocessed HSI are used as reference images, and the SPOT-4 MSI spectral response function is used to obtain the simulated MSI at different times. In addition, Gaussian blur kernel and downsampling factor 4 are used to obtain simulated HSI. In the experiment, all simulated HSIs are 200 × 200 × 100 pixels, simulated MSIs are 800 × 800 × 4 pixels, and reference images are 800 × 800 × 100 pixels. The HSI acquisition dates of simulated dataset-1 are December 19, 2020 and June 5, 2021. The HSI acquisition dates of simulated dataset-2 are December 19, 2020 and June 5, 2021. The HSI acquisition dates of simulated dataset-3 are January 13, 2020 and March 9, 2021 (see Fig. 6).

The real dataset-1 covers part of the Yellow River Estuary. In the experiment, we collect HSI and MSI captured by the ZY-1E satellite, and image preprocessing is performed, including ortho correction, radiometric calibration, atmospheric correction, bad band elimination, and image registration (ZY-1E HSI and MSI are downloaded from https://data.cresda.cn). After the ZY-1E HSI is cut and the bad bands are eliminated, 100 bands of 260 × 260 pixels are used in the experiment. The spatial resolution of ZY-1E MSI is 10 m, including eight bands of 780 × 780 pixels. The final acquisition date of T1 HSI and MSI is September 29, 2021, and T2 MSI is December 19, 2021. The real dataset-2 covers part of Hangzhou Bay. The HSI in the experiment comes from the ZY-1E satellite sensor (ZY-1E HSIs are downloaded from https://data.cresda.cn), and the MSI comes from the Sentinel-2A (S2A) satellite sensor (S2A MSIs are downloaded from https://scihub.copernicus.eu/dhus/#/home). After image preprocessing, image clipping, and bad band elimination, the HSI with 260 × 260 × 100 pixels and MSI with 780 × 780 × 4 pixels are used in the experiment. The final acquisition date of T1 HSI and MSI is January 3, 2021, and T2 MSI is May 17, 2021 (see Fig. 7).

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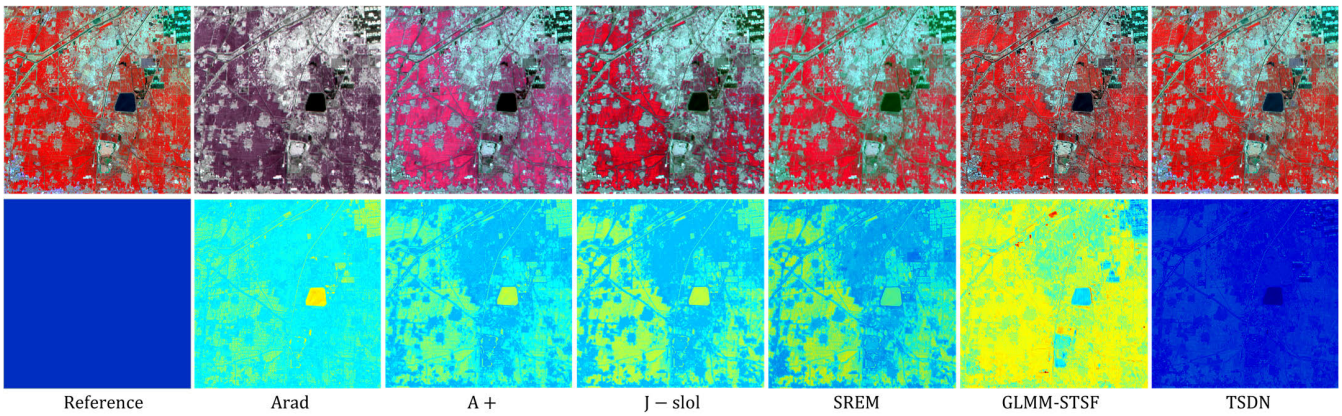


Fig. 8. Experimental results of simulated dataset-1.

TABLE II
QUANTITATIVE RESULTS OF SIMULATED DATASET-1

Methods	Arad	A+	J-slol	SREM	GLMM-STSF	TSDN
SAM	27.0320	25.4323	22.1607	21.9112	31.4990	**4.6971**
ERGAS	25.2913	25.5403	18.9554	17.6945	15.7492	**6.7853**
PSNR	5.8931	7.0652	14.9321	17.7023	15.9647	**30.7039**
CC	0.7054	0.8474	0.7249	0.7692	0.9210	**0.9782**

B. Experimental Setting

For simulated datasets, TSDN randomly clips 20 × 20 HSI and 60 × 60 MSI in each iteration for model training, and 200 × 200 HSI and 800 × 800 MSI are used for model testing. For real datasets, TSDN randomly clips 16 × 16 HSI and 64 × 64 MSI in each iteration for model training, and 260 × 260 HSI and 780 × 780 MSI are used for model testing. In experiments, the number of iterations is set to 10 000, the learning rate is set to 1e-4, Adam with a $\beta_1 = 0.9$, $\beta_2 = 0.99$ is used as the optimizer, and convolution kernel sizes of different networks are shown in Figs. 2–4. Ultimately, all the experiments are performed on a Window10 X64 computer with 64-GB CPU memory and 1 x GPU of GeForce GTX 3060Ti.

In addition, due to the lack of STSF methods for HSI and MSI, four spectral super-resolution methods, e.g., spectral resolution enhancement method (SREM) [56], the hyperspectral sparse reconstruction method proposed by Arad and Ben-Shahar [57], the improved coefficient reconstruction method A+ [58], and J-slol [59], and the STSF method GLMM-STSF, are used for comparison [12].

To quantitatively evaluate fusion results, spectral angle measurement (SAM) [19], Erreur Relative Global Adimini-sionnelle de Synth è se (ERGAS) [9], peak signal-to-noise tone ratio (PSNR) [14], and cross correlation (CC) [7] are used in experiments. SAM is used to evaluate the spectral distortion of the fused image. The larger SAM is, the more serious the spectral distortion is, and the ideal value is 0. CC is used to evaluate the spatial distortion of the image, with an ideal value of 1. ERGAS and PSNR are used to evaluate the comprehensive quality of the fused image. The ideal value of ERGAS is 0, and the larger the PSNR value is, the better the fusion quality is.

C. Experimental Results of Simulated Datasets

Fig. 8 shows the composite false-color fusion results of different methods in the simulated dataset-1. Arad and A+ have obvious spatial details and serious color differences. Compared with the reference image, they cannot accurately predict spectral changes. J-slol is excessively sharp in the vegetation area and has an obvious color difference compared with the reference image in the vegetation area. GLMM-STSF achieves the best visualization results among all benchmark methods. SAM map shows that GLMM-STSF and Arad have the most serious global spectral distortion, and A+, J-slol, and SREM have less spectral distortion in bare land areas. However, serious spectral distortion exists in vegetation and water areas. Compared with other methods, TSDN obtains the best visualization results and SAM map and is the closest to the reference image. Table II shows the quantitative evaluation results of fused images. GLMM-STSF obtains the worst SAM and higher CC, which indicates that GLMM-STSF cannot correctly predict spectral changes and can effectively predict changes of spatial information. Arad, A+, J-slol, and SREM obtain poor quantitative evaluation results, SAM > 20, ERGAS > 17, PSNR > 18, and CC > 0.85, indicating that these methods not only incorrectly predict spectral changes but also could not correctly predict spatial changes. TSDN achieves the best results in all indicators and can correctly predict the spectral and spatial changes.

Fig. 9 shows the fusion results of different methods in the simulated dataset-2. All the benchmark methods have obvious color differences compared with the reference image, so they cannot accurately predict the spectral change, and J-slol and Arad have the worst prediction results in the vegetation area. In addition, Arad, A+, and SREM have fuzzy edge information. The visualization result of GLMM-STSF is superior to all

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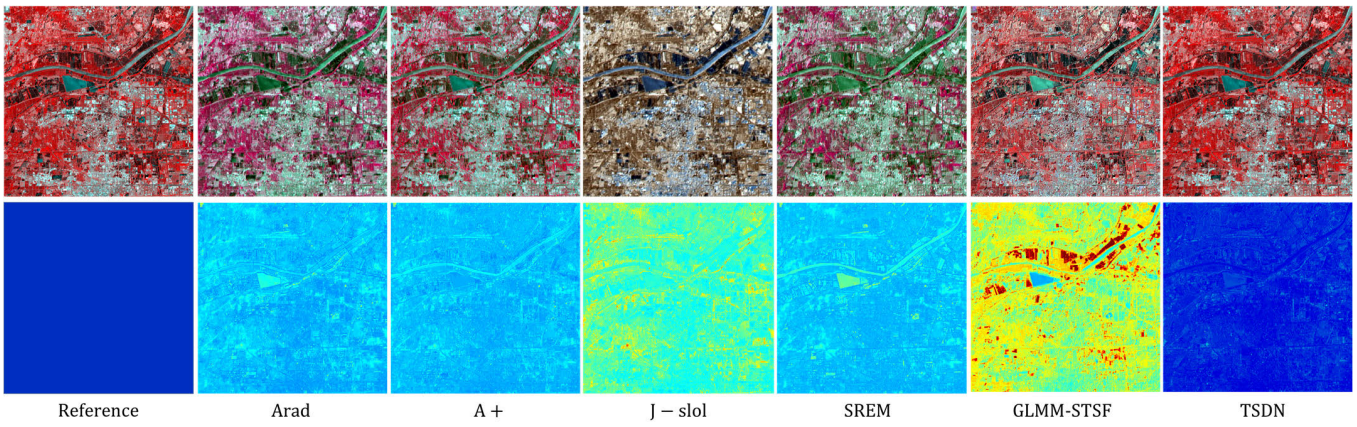


Fig. 9. Experimental results of simulated dataset-2.

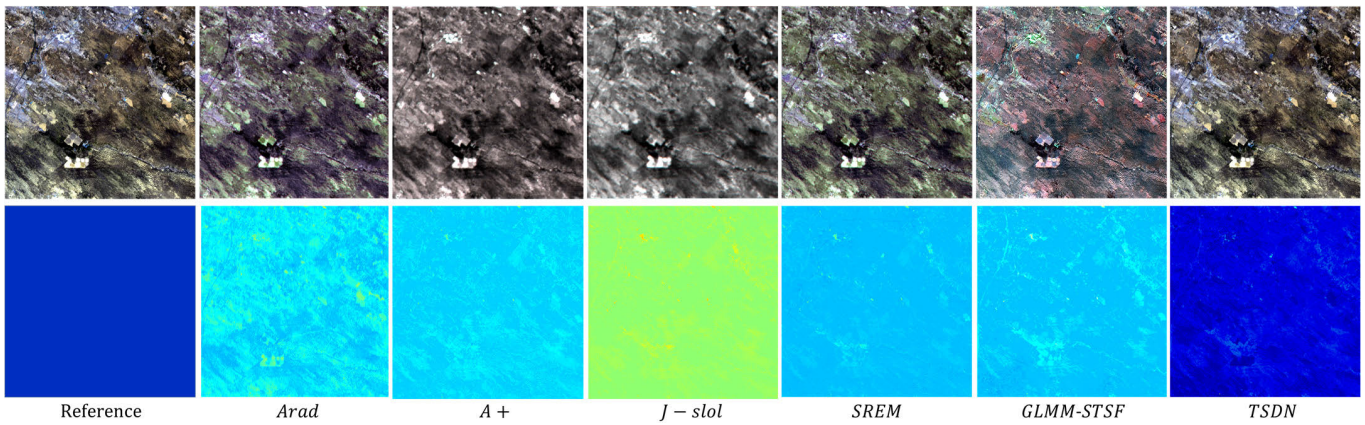


Fig. 10. Experimental results of simulated dataset-3.

TABLE III
QUANTITATIVE RESULTS OF SIMULATED DATASET-2

Methods	Arad	A+	J-slol	SREM	GLMM-STSF	TSDN
SAM	20.6250	18.6484	29.5296	24.6484	32.8763	**4.2169**
ERGAS	25.0875	25.7316	35.1586	15.2642	37.5086	**4.8671**
PSNR	4.3869	8.9459	18.4988	16.0210	16.3494	**32.0946**
CC	0.81548	0.84317	0.8290	0.7825	0.8849	**0.9751**

benchmark methods. SAM maps show that GLMM-STSF and J-slol have the most serious spectral distortion and cannot accurately predict spectral changes. Arad and SREM have more spectral distortion in the water area than in the vegetation area, and A+ has the least spectral distortion in all benchmark methods. Our TSDN still achieves the best visualization results in the simulated dataset-2, with minimal spectral distortion. Table III shows the quantitative evaluation results of different methods. SREM has achieved better SAM and ERGAS, J-slol has achieved better PSNR, and GLMM-STSF has better CC. However, SAM $>$ 14, EGRAS $>$ 15, PSNR $>$ 19, and CC $>$ 0.9 of all benchmark methods indicate that these methods cannot correctly predict spectral and spatial changes. TSDN with SAM $>$ 5, ERGAS $>$ 5, PSNR $>$ 32, and CC $>$ 0.97 is significantly superior to these benchmark methods.

Fig. 10 shows the fusion results of the simulated dataset-3. All benchmark methods have obvious color differences compared with reference image, and the visualization results are poor. GLMM-STSF has better spatial details, followed by Arad and A+, while J-slol and SREM cannot enhance spatial information well. SAM maps show that J-slol and Arad obtain the most serious global spectral distortion. The spectral distortion of A+, SREM, and GLMM-STSF is relatively small, but the global spectral prediction is still inaccurate. The composite false-color image of TSDN is almost consistent with the reference image, and the SAM error is small. Table IV shows the quantitative evaluation results of fused images. TSDN still achieves the best value in all indicators. J-slol has the worst SAM, A+ achieves the worst ERGAS, and SREM has a better PSNR in all benchmark methods, but the PSNR is only 18.8712. GLMM-STSF always has advantages in spatial prediction, and CC $>$ 0.84.

Fig. 11 shows the PSNR and CC of each band of the fused image. The PSNR of all bands of TSDN fusion result

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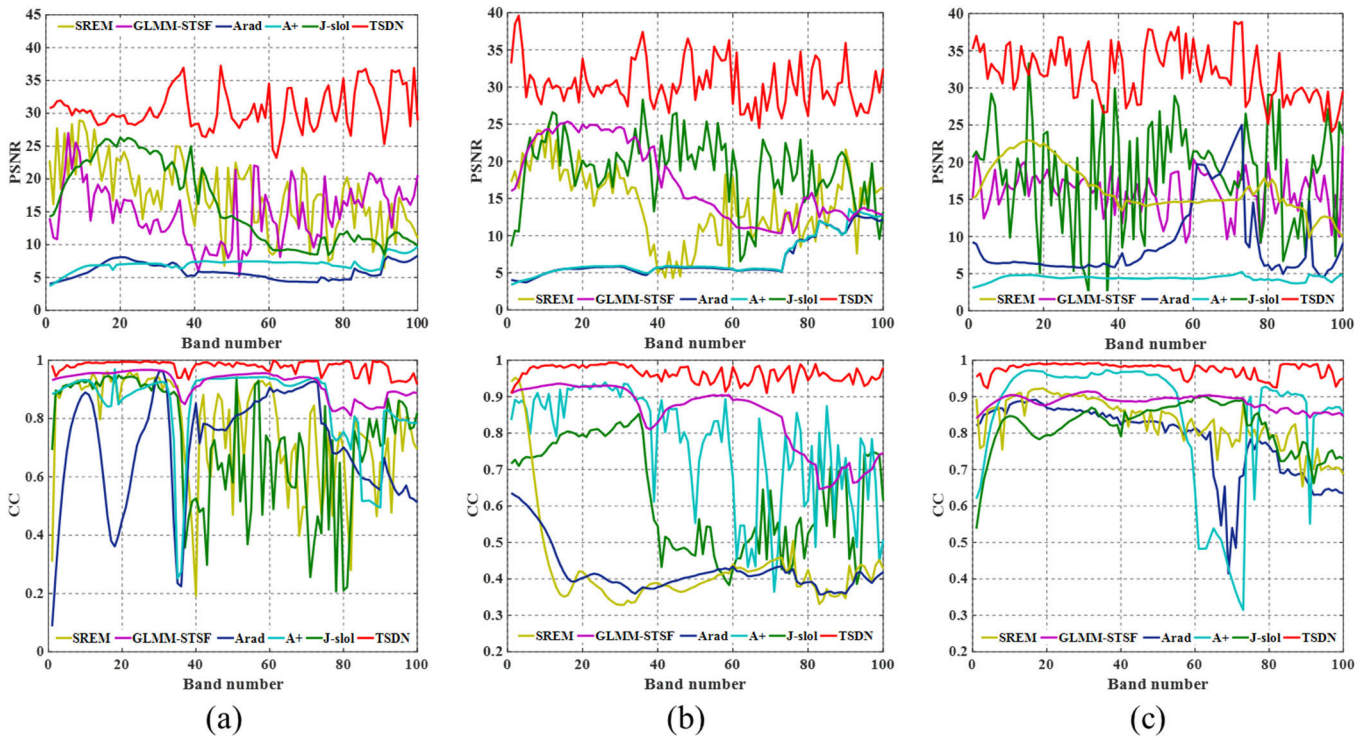


Fig. 11. CC and PSNR of each band of fusion results. (a) Simulated dataset-1, (b) simulated dataset-2, and (c) simulated dataset-3.

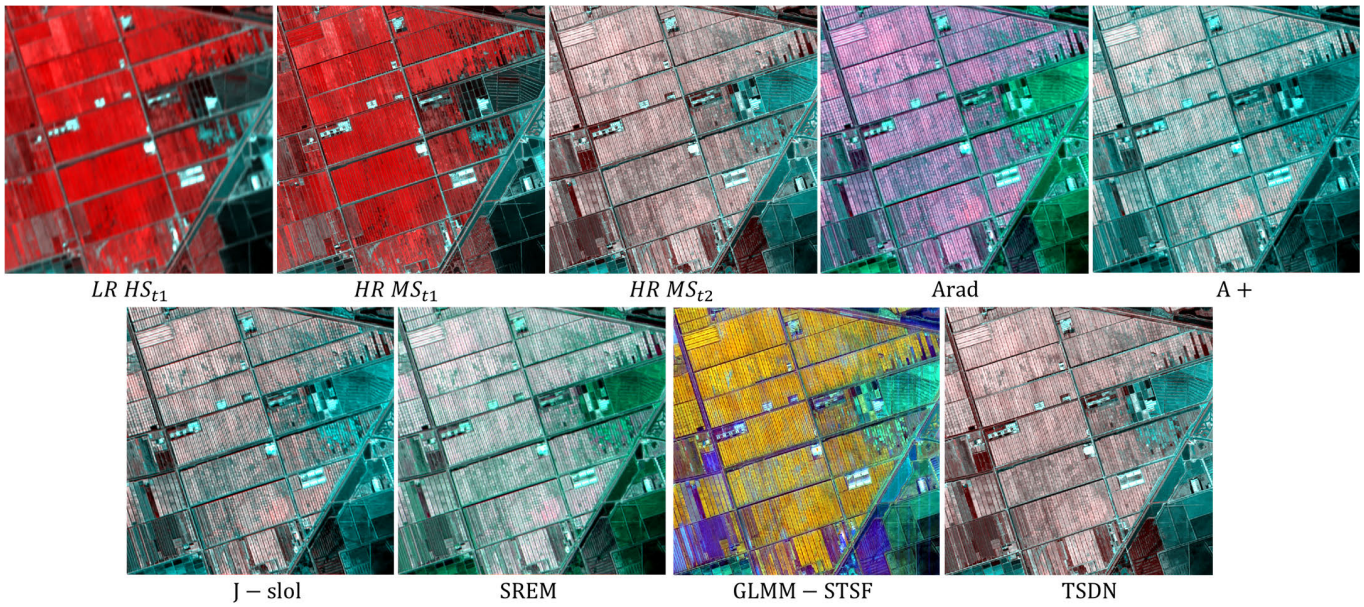


Fig. 12. Experimental results of real dataset-1.

TABLE IV
QUANTITATIVE RESULTS OF SIMULATED DATASET-3

Methods	Arad	A+	J-slol	SREM	GLMM-STSF	TSDN
SAM	26.6845	11.4897	32.4196	11.4106	17.7629	**5.9783**
ERGAS	25.3364	25.3380	17.8754	27.6464	15.935	**7.0161**
PSNR	6.9593	6.8069	14.1812	18.8712	17.1709	**30.5051**
CC	0.4215	0.4306	0.6279	0.7691	0.8467	**0.9636**

is >23, and CC > 0.9. In the fusion results of J-slol, SREM, and GLMM-STSF, the PSNR between bands fluctuates significantly. Arad and A+ achieve the lowest PSNR in almost all bands. In the fusion results of J-slol and A+, the CC between bands fluctuates seriously. GLMM-STSF achieves better CC values in all bands. Only some bands of SREM and

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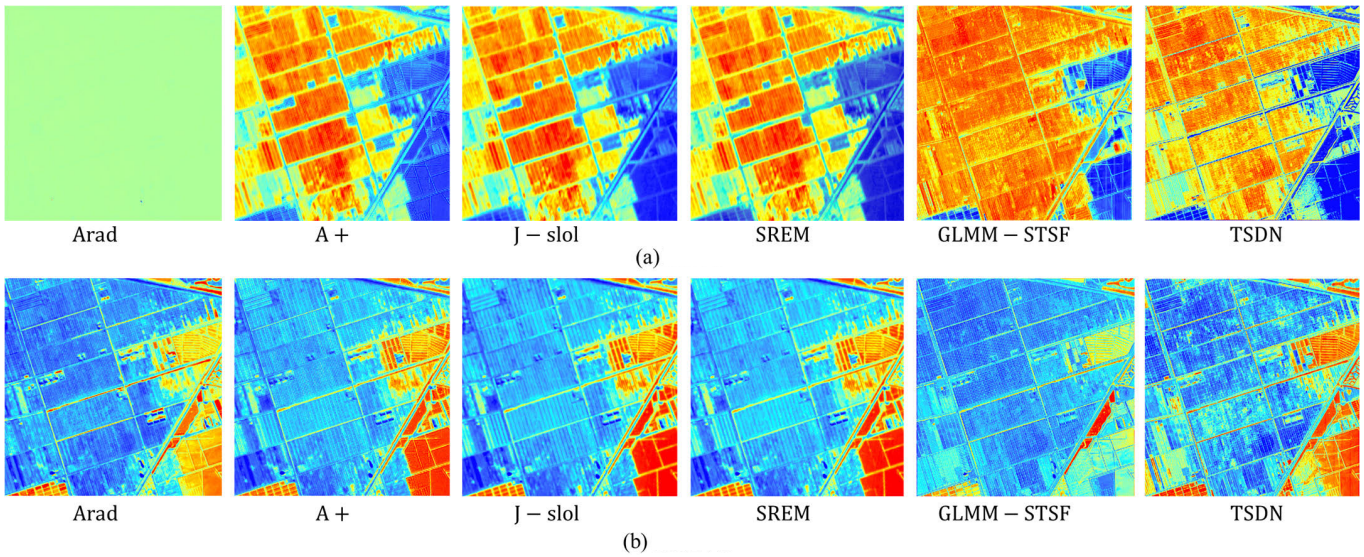


Fig. 13. Index calculation results of fused images in real dataset-1. (a) NDVI. (b) MNDWI.

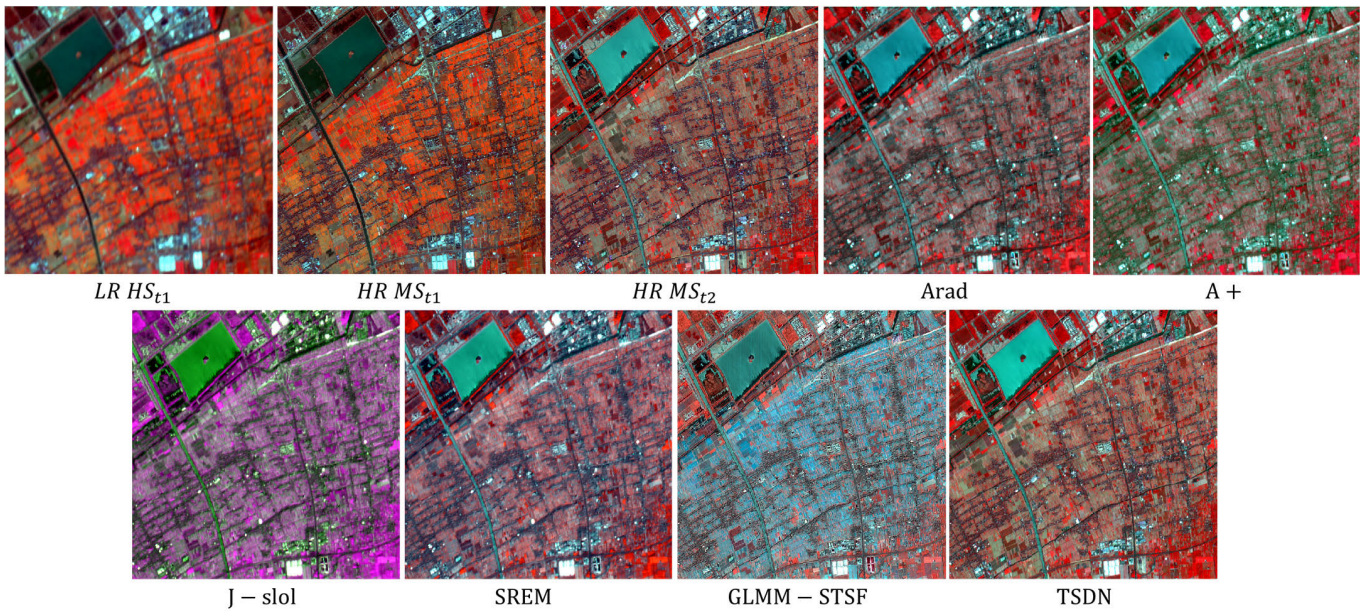


Fig. 14. Experimental results of real dataset-2.

Arad achieve better CC, and CC of numerous bands is less than 0.8.

D. Experimental Results of Real Datasets

Fig. 12 shows the fusion results of different methods in the real dataset-1. All the methods obtain obvious spatial details. However, there are obvious spectral differences between the fused images. Arad and GLMM-STSF obtain the worst visualization results and did not correctly predict the change of spectral information. Although A+, J-slol, and SREM are close to T2 MSI, there are always serious spectral errors. TSDN achieves the best visual results. Due to the lack of reference images, we calculated the NDVI and modified normalized difference water index (MNDWI) of different fusion images. Fig. 13 shows the index calculation results. The NDVI of Arad is completely wrong, and A+, J-slol, and SREM obtain similar NDVI results. However, vegetation is overenhanced and some vegetation is not identified. The NDVI of GLMM-STSF is excessively enhanced, and some nonvegetated areas are misidentified as vegetated areas. TSDN can correctly identify the vegetation area and achieve the best results. For MNDWI, Arad weakens MNDWI in some areas, A+, J-slol, and SREM obtain similar MNDWI, but some water body areas are not recognized. GLMM-STSF not only weakens MNDWI but also a large number of water areas are not identified. TSDN still achieves the best MNDWI.

Fig. 14 shows the fusion result of real dataset-2. The fusion results of all methods have obvious spatial details. However, the spectral differences are large. J-slol and GLMM-STSF have the worst visualization results. Arad, A+, and SREM have better visualization results in densely vegetated areas, but poor visualization results in sparsely vegetated areas. TSDN

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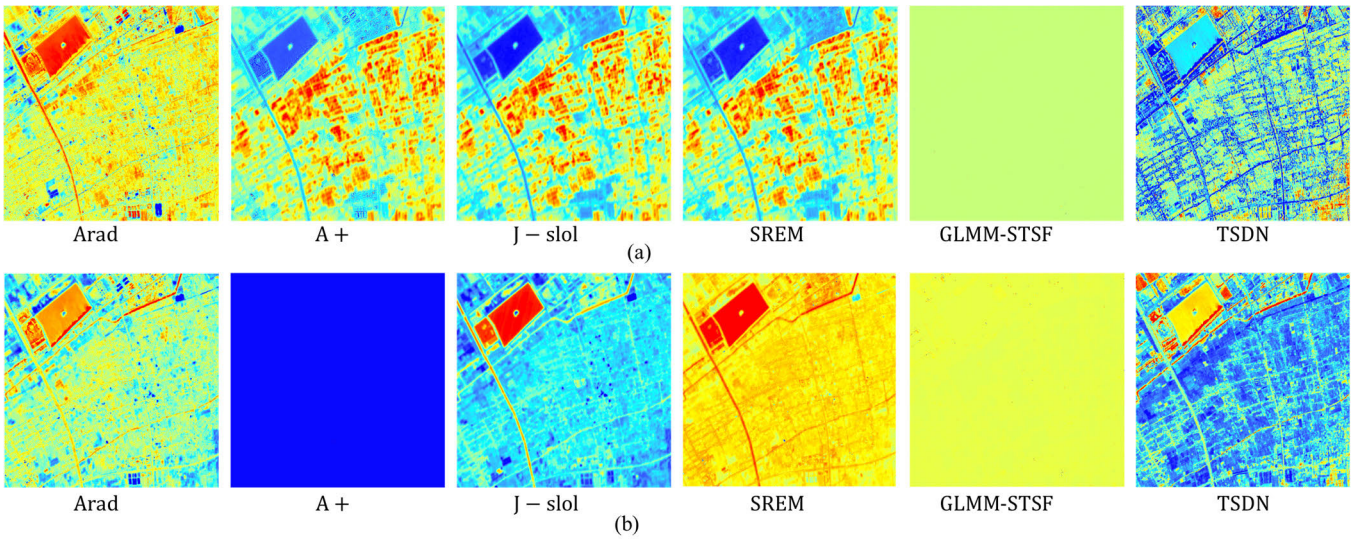


Fig. 15. Index calculation results of fused images in real dataset-2. (a) NDVI. (b) MNDWI.

achieves the best visualization results. Fig. 15 shows the index calculation results of the fused images. GLMM-STSF cannot calculate NDVI. Arad identifies water as vegetation, and NDVI is completely wrong. A+, J-slol, and SREM obtain approximate NDVI results. However, these methods are overenhanced for some sparsely vegetated areas. TSDN obtains the best NDVI calculation results. In addition, A+ and GLMM-STSF cannot calculate MNDWI. Arad and SREM identify a large amount of vegetation as water. J-slol achieves relatively correct results among all benchmark methods. TSDN obtains the best MNDWI.

E. Discussion

TSDN obtains comparative STSF results of HSI and MSI in the simulated datasets and the real datasets. The main reasons are given as follows.

1) TSDN introduces a subspace extraction network, which reduces the modal difference between images. The difference information of T1 and T2 MSI subspaces is used to inject T1 HSI subspaces, which accurately predicts the subspaces of T2 HSI.
2) TSDN introduces tensor decomposition theory, extracts image features from spatial-x, spatial-y, and spectral-z dimensions, and predicts the change information of the three dimensions, which not only improves the prediction accuracy of spectral changes but also enhances the prediction accuracy of spatial changes.
3) TSDN designs a reasonable loss function, which constrains the tensor subspace extraction network and TSN and improves the training performance of the network.

We have verified the importance of 3-D convolution in TSDN. For TSDN without multidirectional convolution, we uniformly set the convolution kernel to 3×3 and set the same epoch to 10 000 in the experiment. Fig. 16 shows the fusion results, and 3-D convolution can reconstruct the fused image more accurately.

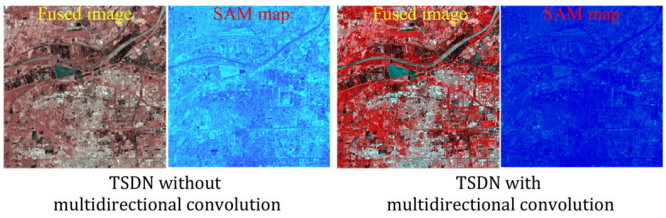


Fig. 16. Experimental results of 3-D convolution ablation.

TABLE V
QUANTITATIVE RESULTS OF ABLATION EXPERIMENT OF TENSOR DECOMPOSITION NETWORK

Tensor decomposition network	SAM	CC
×	12.3642	0.8138
√	4.6971	0.9782

To verify the importance of subspace extraction, we show the feature maps of TSDN without loss_{tse}^{t1} and the feature maps of TSDN with loss_{tse}^{t1} in Fig. 17. It is clear that loss_{tse}^{t1} forces the network to learn common subspace features and reduces modal differences between data. Fig. 15 also shows the influence of subspace extraction on STSF results. The results show that the subspace extraction network greatly improves the spectral and spatial prediction accuracy. It is worth noting that TSDN uses different information between T1 and T2 MSI tensor subspaces to inject into T1 HSI tensor subspaces. Therefore, the number of features between images must be consistent, and the introduction of subspace is necessary.

We also verify the influence of tensor decomposition network on TSDN by using 10 000 iterations. Fig. 18 and Table V show the fusion results of TSDN with tensor decomposition network and TSDN without tensor decomposition network. The experimental results show that tensor decomposition network is very important for TSDN. When there is no

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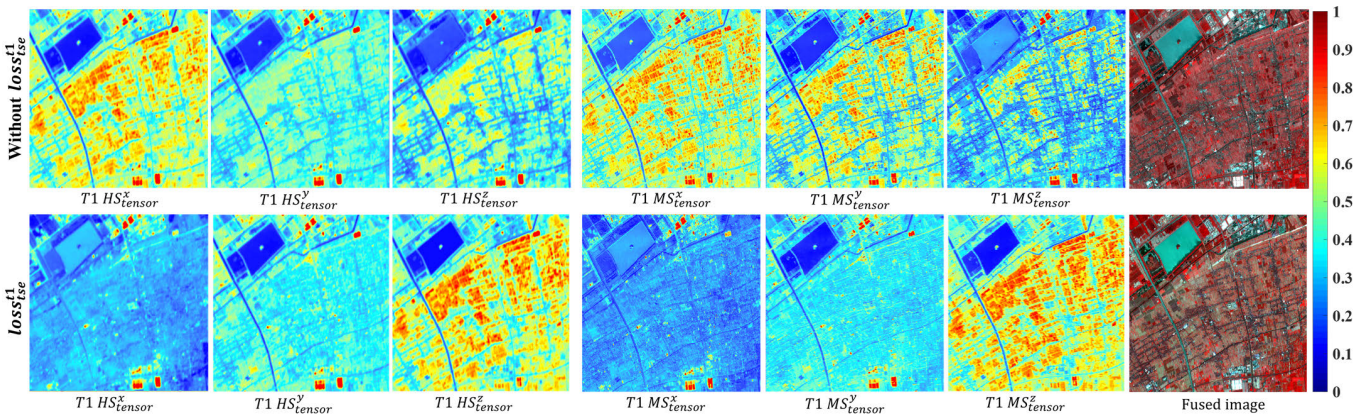


Fig. 17. Results of ablation experiment of subspace extraction network.

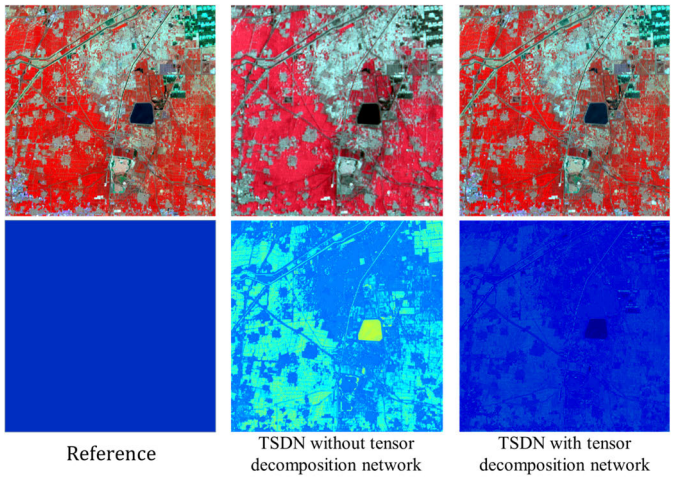


Fig. 18. Results of ablation experiment of tensor decomposition network.

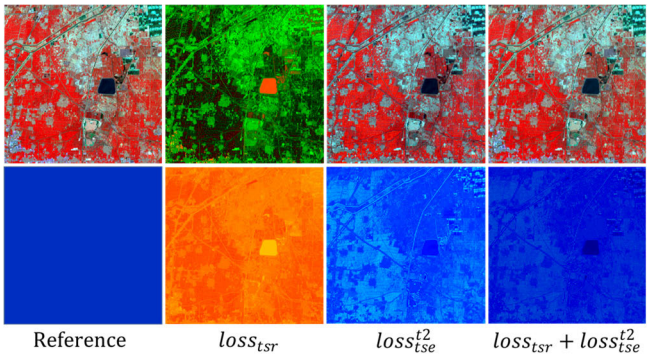


Fig. 19. Results of ablation experiment of loss.

TABLE VI
QUANTITATIVE RESULTS OF ABLATION EXPERIMENT OF LOSS

$loss_{tsr}$	$loss_{tse}^{t2}$	SAM	CC
×	√	25.6981	0.7795
√	×	12.3456	0.8469
√	√	4.6971	0.9782

TABLE VII
CALCULATION TIME FOR DIFFERENT METHODS

Methods	Computing time (min)	
	Real dataset-1	Real dataset-2
Arad	36.32	38.15
A+	31.57	39.19
J-slol	50.19	48.11
SREM	42.50	45.41
GLMM-STSF	89.14	93.36
TSDN	63.45	59.23

tensor decomposition network, TSDN has obvious spectral and spatial prediction errors, and SAM $>$ 10 and CC $<$ 0.85. On the one hand, tensor decomposition network can extract more abundant feature information and improve the network training performance. On the other hand, it can predict missing data from spatial-x, spatial-y, and spectral-z dimensions, which can effectively constrain the change loss between images.

We perform ablation experiments on loss_{tsr} and loss_{tse}^{t2}. loss_{tsr} ensures that the fusion image and T2 MSI have the same tensor subspaces, and loss_{tse}^{t2} ensures that the tensor subspaces of T1 HSI can effectively reconstruct T1 HSI. The combination of loss_{tsr} and loss_{tse}^{t2} guarantees that the fused image has the correct spectral and spatial information. To verify the importance of loss_{tsr} and loss_{tse}^{t2}, Fig. 19 and Table VI show the fusion results of loss_{tsr}, loss_{tse}^{t2}, and the combination of both used. The experimental results show that the fusion result of loss_{tsr} cannot correctly predict the T2 HSI, SAM $>$ 20, and CC $<$ 0.8. The fusion result of loss_{tse}^{t2} is better than that of loss_{tsr}, but the obvious error still exists. $\text{loss}_{tsr} + \text{loss}_{tse}^{t2}$ can effectively improve the prediction accuracy of fusion results in limited iterations.

Finally, we compared the computational efficiency of different methods. Table VII presents the computation time of various methods on real datasets. Arad, A+, J-slol, and SREM exhibit higher computational efficiency because these methods only operate on HSI. However, GLMM-STSF and TSDN require the fusion of HSI and MSI information. It is worth noting that TSDN demonstrates superior computational

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efficiency compared to GLMM-STSF, due to its weight sharing approach, which significantly reduces the number of trainable parameters. In contrast, GLMM-STSF employs ADMM optimization, which involves a substantial number of unknown parameters that need to be optimized.

IV. Conclusion

Remote sensing images with high temporal, spatial, and spectral resolution can be obtained by fusion of MSI and HSI. TSDN, a novel, robust HSI, and MSI spatial–temporal–spectral fusion method, is proposed in this article. TSDN establishes the 3-D HSI tensor subspace extraction network and the 3-D MSI tensor subspace extraction network through joint training to obtain the tensor subspace features of upsampled T1 HSI and Gaussian blurred T1 MSI, which reduced the modal difference between data. Then, the trained 3-D MSI tensor subspace extraction network is mapped to the Gaussian blurred T2 MSI to obtain the Gaussian blurred T2 MSI tensor subspace features. After that, the change information is obtained by subtracting the Gaussian blurred T2 MSI tensor subspace features from the Gaussian blurred T1 MSI tensor subspace features, and the change information is injected into the upsampled T1 HSI tensor subspace features to obtain the T2 HSI tensor subspace features with low spatial resolution. Finally, TSDN established a 3-D HSI TSN to reconstruct T2 HSI by taking T2 HSI tensor subspace features with low spatial resolution and T2 MSI as input data. The fusion performance of TSDN has been verified by using simulated datasets and real datasets, and we make the following conclusions.

1) TSDN achieves satisfactory fusion performance and is robust and practical.
2) The fusion performance of TSDN is better than that of Arad, A+, J-slol, SREM, and GLMM-STSF. TSDN accurately predicts spectral changes and spatial changes.
3) The introduction of tensor subspace theory significantly improves the fusion performance of TSDN, and the loss function we designed ensures the unsupervised training of the network.

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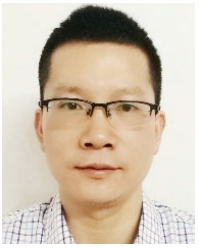
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Weiwei Sun (Senior Member, IEEE) received the B.S. degree in surveying and mapping and the Ph.D. degree in cartography and geographic information engineering from Tongji University, Shanghai, China, in 2007 and 2013, respectively.

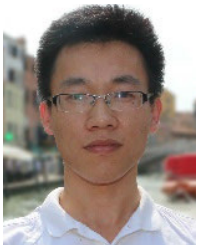
From 2011 to 2012, he was a Visiting Scholar with the Department of Applied Mathematics, University of Maryland at College Park, College Park, MD, USA, with the famous professor John Benedetto to study on the dimensionality reduction of hyperspectral image. From 2014 to 2016, he was a Post-Doctoral Fellow with the State Key Laboratory for Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China, to study intelligent processing in hyperspectral imagery. From 2017 to 2018, he was a Visiting Scholar in hyperspectral image processing with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. He is currently a Full Professor with Ningbo University, Ningbo, China. He has published more than 80 journal papers and his research interests include hyperspectral image processing with machine learning.



Kai Ren (Student Member, IEEE) received the B.S. degree in humanistic geography and urban–rural planning from Shanxi Normal University, Xi'an, China, in 2018, and the M.S. degree in humanistic geography from Ningbo University, Ningbo, China, in 2021. He is currently pursuing the Ph.D. degree with the Faculty of Electrical Engineering and Computer Science, Ningbo University.

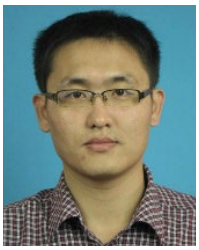
His research interests include remote sensing data processing and data analysis.

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Xiangchao Meng (Member, IEEE) received the B.S. degree in geographic information system from Shandong University of Science and Technology, Qingdao, China, in 2012, and the Ph.D. degree in cartography and geography information system from Wuhan University, Wuhan, China, in 2017.

He is currently an Associate Professor with the Faculty of Electrical Engineering and Computer Science, Ningbo University, Ningbo, China. His research interests include remote sensing image fusion and quality evaluation.



Gang Yang received the M.S. degree in geographical information system from the Hunan University of Science and Technology, Xiangtan, China, in 2012, and the Ph.D. degree from the School of Resource and Environmental Sciences, Wuhan University, Wuhan, China, in 2016.

He is currently an Associate Professor with Ningbo University, Ningbo, China. His research interests focus on missing information reconstruction of remote sensing images, cloud removal of remote sensing images, and remote sensing time-series products temporal reconstruction.



Jiangtao Peng (Senior Member, IEEE) received the B.S. and M.S. degrees from Hubei University, Wuhan, China, in 2005 and 2008, respectively, and the Ph.D. degree from the Institute of Automation, Chinese Academy of Sciences, Beijing, China, in 2011.

He is currently a Professor with the Faculty of Mathematics and Statistics, Hubei University. His research interests include machine learning and hyperspectral image processing.



Jiancheng Li received the B.S., M.S., and Ph.D. degrees from the Wuhan Technical University of Surveying and Mapping, Wuhan, China, in 1987, 1990, and 1993, respectively.

From 1994 to 1995, he was a Visiting Scholar with the University of Texas Center for Space Research, Austin, TX, USA, and a Senior Visiting Scholar with the Finland National Institute of Geodesy, Finland. From 2001 to 2001, he was a Senior Visiting Scholar with the German Research Center for Geosciences (GFZ), Potsdam, Germany. From 2002 to 2003, he was a Researcher with the German Research Center for Geosciences (GFZ), Potsdam. From August 2008 to September 2008, he was a Senior Visiting Scholar with Germany Stuttgart University, Stuttgart, Germany. He is currently a Full Professor with the Faculty of Surveying and Mapping, Wuhan University, Wuhan. His research interests include geodesy and remote sensing.

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